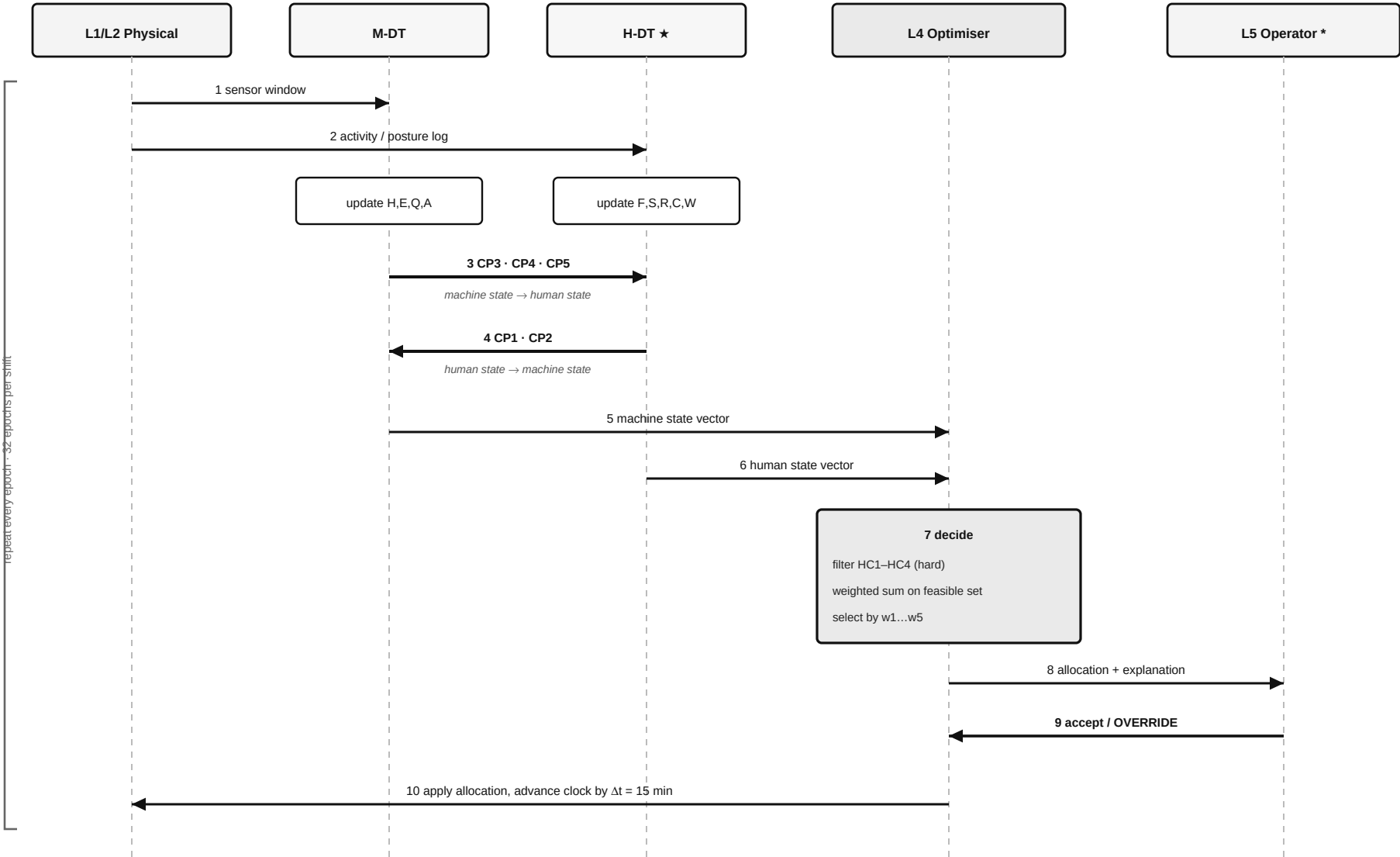


Twin Coupling and Decision Flow within One 15-minute Epoch



Coupling points — the mechanism absent from static formulations

CP1	skill → quality	defect risk rises as skill falls: term $\beta_2(1 - S_{h,t})$ (normalised) in Q_m	CP1, CP2: H-DT → M-DT
CP2	fatigue → quality	a tired operator produces more scrap: term $\beta_3 F_h$ (normalised) in Q_m	CP3, CP4, CP5: M-DT → H-DT
CP3	health → cognition	a degraded machine demands attention: term $\gamma_1(1 - H_m)$ in C_h	Removing all five reduces the framework to two independent static parameter sets.
CP4	intensity → fatigue	heavier task raises the fatigue asymptote: $E'_w(\text{task})$ in F_h	
CP5	speed → ergonomics	faster machine worsens posture / safety margin: term $\psi_2 v_m$ (normalised) in R_h	

* L5 is specified in the architecture but is not implemented in this study (steps 8 and 9).